Daniele Visioni

Curriculum Vitae

Education

2020-2021 **Postdoc Leadership Program**, *Cornell University*, Ithaca (NY).

2015-2018 **Ph.D. with Honors in Atmospheric Physics and Chemistry**, *University of L'Aquila*, Italy.
Thesis: A climate engineering technique for a warming planet: stratospheric sulfur injection as a temporary solution to greenhouse gases increase.

2013-2015 **Master's Degree in Physics**, *University of L'Aquila*, Italy, Curriculum in Atmospheric Physics.

2009-2013 **Bachelor's Degree in Physics**, *University of L'Aquila*, Italy.

Professional appointments

August 2025-Ongoing **Head of Data**, *Reflective*.
Leadership role for data-related products, including the development of a open-source, free CloudHub resource for SRM research (`https://reflective.2i2c.cloud`), the Simulator (`https://simulator.reflective.org/`) and a Multi-Scale Modeling Project

June 2024-August 2025 **Scientific Advisor**, *Reflective*.
Help develop overall strategy, advise on request for proposals and co-lead development of the Simulator

2023-Present **Cornell Atkinson Faculty Fellow**, *Cornell Atkinson Center for Sustainability*, Ithaca (NY), USA.

August 2023-Present **Assistant Professor**, *Cornell University - Department of Earth and Atmospheric Sciences*, Ithaca (NY), USA.

Jan 2023-July 2023 **Project Scientist I**, *National Center for Atmospheric Research - Atmospheric Chemistry, Observation and Modeling Lab*, Boulder (CO), 40% Part-time position.

2022-2023 **Research Associate**, *Cornell University - Sibley School of Mechanical and Aerospace Engineering*, Ithaca (NY), USA.

2018-2022 **Post-doctoral Associate**, *Cornell University - Sibley School of Mechanical and Aerospace Engineering*, Ithaca (NY), USA, Supervisor: Prof. Douglas MacMartin.

2015-2018 **Ph.D. Fellow in Atmospheric Physics and Chemistry**, *University of L'Aquila, Italy*, Supervisor: Prof. Giovanni Pitari.

Jan-Mar 2018 **Visiting Scientist**, *NCAR*, Boulder (CO), USA, Supervisor: Dr. Simone Tilmes.

June-Sep 2017 **Visiting Scientist**, *NASA GSFC - Earth Science Division*, Greenbelt (MD), USA.

Awarded Research Grants

2026-2028 **National Oceanic and Atmospheric Administration - Climate Program Office**, *Benchmarking climate models' behavior for Stratospheric Aerosol Injections applications to narrow uncertainties*, Recommended for Funding, pending availability: 880.000$, PI: D. Visioni, co-PIs: E. Bednarz, I. Quaglia, D. Watson-Parris.
Role: overall ideation and coordination of the project

2025-2026 **Bernard and Anne Spitzer Charitable Trust**, *SRM Downscaling Project with CarbonPlan*, Awarded, 100.000$, PIs: D. Visioni, O. Chedwigg.
Role: data gathering and validation of downscaled datasets

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2025-2027 **Advanced Research+Innovation Agency (UK)**, *Planetary Sunshade Baseline Survey*, Awarded, 274.974$, PIs: Morgan Goodwin (Planetary Sunshade Foundation), D. Visioni.
Role: climate modeling of different solar shading architectures

2025-2027 **Advanced Research+Innovation Agency (UK)**, *Global to Local Impacts of SRM Project (GLISP)*, Awarded, 270.000$, PIs: The DEGREES Initiative (UK), D. Visioni, Chris Lennard (University of South Africa).
Role: testing of different downscaling strategies, and data availability

2025-2027 **Advanced Research+Innovation Agency (UK)**, *Defining the minimum scale of an SAI test*, Awarded, 575.348$, PIs: D. Visioni, D.G. MacMartin.
Role: identifying volcanic eruptions to use as proxies for SAI tests

2024-2029 **Quadrature Climate Foundation**, *Geoengineering Assessment Across Uncertainties, Scenarios and Strategies (GAUSS)*, Awarded, 5.000.000$, PIs: D.G. MacMartin, D. Visioni.
Role: overall ideation and coordination of the project

2024-2025 **Quadrature Climate Foundation**, *Supporting GeoMIP activities in 2024-2025*, Awarded, 250.000$, PI: D. Visioni.
Role: general GeoMIP coordination and disbursement of travel grants

2023 **Cornell Atkinson Center for Sustainability - 2023 Fast Grant Awards**, *Changes in Shipping Emissions as a Natural Analogue for Climate Intervention: detecting and attributing changes due to specific human activities as a testbed for future controversies.*, Awarded, 23.050 $, PI: D. Visioni.
Role: overall ideation and coordination of the project

2022 **National Oceanic and Atmospheric Administration - Earth Radiation Budget Program**, *Atmospheric Aerosols and their Potential Roles in Solar Climate Intervention Methods grant - Assessing the impact of Stratospheric Aerosol Interventions on Regional Climate and Air Quality*, Awarded (NA22OAR4310477), PI: S. Tilmes, listed as co-PI.
Role: ran CESM2 simulations for future intercomparisons

2022 **Resources for the Future - Social Science Research into Solar Geoengineering**, *Integrating Risk Perception with Climate Models to Understand the Potential Deployment of Solar Radiation Modification to Mitigate Climate Change*, Awarded, 10.000 $, PIs: B. Beckage, D. Visioni.
Role: providing climate and SRM expertise to the integration

Scientific Leadership and Service Activities

September 2025-Ongoing **Member of the Intervention Risk Review Group**, *Reef Restoration and Adaptation Program (RRAP), Australia.*
In an advisory capacity, nominated by the RRAP board to identify risks, review relevant output by RRAP and provide independent advice `https://gbrrestoration.org/our-team/advisory-and-working-groups/intervention-risk-review-group/`.

August 2025-Ongoing **Intergovernmental Panel on Climate Change**, *Lead Author, Ch. 9, WG1 "Earth System Responses under Pathways Toward Temperature Stabilization, including Overshoot Pathways.*

August 2024 **Consultant for the U.S. Department of State on Climate Technology**.

2024-2028 **Gordon Research Conference on Climate Engineering**, *Vice-Chair of the 2026 meeting, Chair of the 2028 meeting*, `https://www.grc.org/climate-engineering-conference/2026/`.

January 2024-Ongoing **World Climate Research Program**, *Co-chair of the Lighthouse Activity on Climate Intervention Research*, `https://www.wcrp-climate.org/ci-overview`.

2023-2025 **CMIP Data Request Team**, `https://gmd.copernicus.org/articles/18/2639/2025/`.
Connecting with different communities and interfacing with CMIP to create lists of data requests for modeling teams for CMIP7

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October 2022-Ongoing	**National Center for Atmospheric Research (NCAR)**, *External co-chair for the Whole Atmosphere Working Group (WAWG)*, Assisting and overseeing model development for the Whole Atmosphere Community Climate Model (WACCM) together with the internal NCAR co-chairs.
October 2022-March 2023	**American Geophysical Union (AGU)**, *Expert on panel tasked with updating the Union statement on Climate Intervention*, `https://www.agu.org/Science-Policy/Position-Statements/Climate-Intervention`.
June 27-28, 2022	**Gordon Research Seminar on Climate Engineering**, *Co-chair*, Sunday River-Newry, ME, USA, Early Career conference organization, speakers selection, funding.
2022	**World Climate Research Programme (WCRP) Climate Intervention Task Team**, *Establishing a strategy as to how WCRP can address Climate Intervention research in the future*.
March 2021-July 2022	**WMO - Scientific Assessment of Ozone Depletion 2022**, *Co-author*. Leading Section 3 - "Dynamical and Chemical changes" on Chapter 6: Stratospheric aerosol intervention and its potential effect on the stratospheric ozone layer. Invited to attend the Panel Review Meeting in Geneva to draft the Executive Summary.
Feb 2021-Ongoing	**NCAR HPC User Group Advisor**, *National Center for Atmospheric Research*, `https://www2.cisl.ucar.edu/user-support/ncar-hpc-user-group`. High Performance Computing User Group Advisor at the Computational and Information Systems Lab
March 2021-Ongoing	**Solar Radiation Management Governance Initiative (now Degrees)**, *Research Collaborator*. External Advisor for multiple research teams awarded by SRMGI/Degrees
2019-2025	**AGU Fall Meeting**, *Session Chair or Session Convener on Climate Engineering related sessions*, Multiple locations.
Dec 2020-Ongoing	**EGUsphere Moderator**, *European Geophysical Union*, `www.egusphere.net/`. Moderator for the not-for-profit scientific repository of the EGU, bringing together all preprints submitted to EGU journals.
Aug 2020-Ongoing	**Project Co-Chair**, *Geoengineering Model Intercomparison Project*, `geomip.org`. Coordinating modeling groups, devising modeling experiments, organizing GeoMIP meetings, liaising with WCRP and CMIP, as well as other external groups.

Scholarships and Awards

March 2022	**2022 Future Leader fellow for the Aspen Institute**.
May 2021	**Selected for ACCESS-XVI**, *Atmospheric Chemistry Colloquium for Emerging Senior Scientists*.

Mentorship

(Violet is used for a mentored Grad Student, brown an Undergrad Student or blue for a Postdoc. Their publications are similarly color-coded below.)

Undergraduates, master students, graduate students and postdocs primarily mentored by me					
Name	Role	Location	Period	Accomplishments	Where are they now
Ilaria Quaglia	Grad Student + Postdoc	Univaq (Italy) Cornell (EAS)	2019-2023 2023-2024	2 pubs (ACP) 2 pubs (GRL, ESD)	NCAR Project scientist
Kion Yaghoobzadeh	Undergrad Honors thesis	Cornell (EAS)	2023-2024	Presented at AMS winter meeting	Research associate at Reflective
Illeana Gomez-Leal	Postdoc	Cornell (EAS)	2024-2026	2 preprints (AA)	
James Park	Grad Student	Cornell (EAS)	2024-Curr		Current
Isabelle Amacker	Grad Student	Cornell (EAS)	2024-Curr		Current
Kelsey Roberts	Postdoc	Cornell (EAS)	2024-Curr	1 pub (RevGeo)	Current
Ronald Geiger	Honor thesis + MEng	Cornell (EAS)	2024-Curr		Current
Cindy Wang	Postdoc	Cornell (EAS)	2025-Curr	1 pub (ACP) 1 preprint (EF)	Current IPCC Contrib. A
Brendan Clark	Postdoc	Cornell (EAS)	2025-Curr	1 Pub (GRL)	Current
Endre Farago	Postdoc	Cornell (EAS)	2025-Curr		Current
Temitope Egbebiyi	Postdoc	Cornell (EAS)	2025-Curr	1 preprint (ESD)	Current
Yue Wang	Grad Student	Cornell (EAS)	2025-Curr	3 preprints (JAMES, GRL, npjCA)	Current
Jingwen Lyu	Grad Student	Cornell (EAS)	2025-Curr		Current
Oscar Wang	Undergrad	Cornell (BEE)	2026-Curr		Current

Undergraduates, graduate students and postdocs co-mentored with others					
Name	Role	Location	Period	Achievements	Where are they now
Walker Lee	Grad Student	Cornell (MAE)	2018-2023	4 pubs (ESD, GRL, EF)	NCAR Project scientist
Yan Zhang	Grad Student	Cornell (MAE)	2019-2023	2 pubs (ESD)	Private sector
Jared Farley	Grad Student	Cornell (MAE)	2021-2026	2 pubs (ER:C, GMD)	Scripps Postdoc
Ezra Brody	Grad Student + Postdoc	Cornell (MAE)	2021-2026	3 pubs (ER:C, ESD)	Cornell Postdoc
Lee Webster	Grad Student	Cornell (EAS)	2026-Curr		Current

UnivAQ:University of L'Aquila; **EAS**: Department of Earth and Atmospheric Sciences; **MAE**: Sibley School of Mechanical and Aerospace Engineering; **NCAR**: National Center for Atmospheric Research (Boulder, CO).

Publications

1^{st} author pubs: 19 (year **bolded**); Pubs for which I was primary/co-primary mentor: 18 (year color-coded); h-index: 38 (Scopus)

107. 2026 **On the detectability by uninvolved parties of covert stratospheric aerosol injection programs producing transboundary impacts**, *Smith, W., Alberola, M., Boers, J. G., Rosenlof, K. H.,* ***Visioni, D.***, Environ. Res.: Climate 5 035028. doi:10.1088/2752-5295/ae7148.

106. 2026 **Land models likely underestimate the impact of future atmospheric dryness on European tree growth**, *Clark, B.*, ***Visioni, D.***, *Kothari, S., Lerdau, M.*, Geophysical Research Letters, 53, e2026GL122759. https://doi.org/10.1029/2026GL122759.

105. 2026 **Multi-model analysis of the impact of water vapor on the radiative forcing of volcanic aerosols after the 2022 Hunga Eruption**, *Quaglia, I.*, ***Visioni, D.*** *et al.*, Atmos. Chem. Phys., 26, 7677–7704, https://doi.org/10.5194/acp-26-7677-2026.

104. 2026 **G6-1.5K-SAI and G6sulfur: changes in impacts and uncertainty depending on stratospheric aerosol injection strategy in the Geoengineering Model Intercomparison Project**, *Lee, W. R.*, ***Visioni, D.*** *et al.*, Atmos. Chem. Phys., 26, 7463–7483, https://doi.org/10.5194/acp-26-7463-2026.

103. 2026 **Differentiating solar radiation modification field experiments: Scale, technical characteristics, and governance implications**, *Redmond Roche, B. H., Hernandez-Galindo, I., Määttänen, A., Moore, J. C., Boucher, O., Dollner, B. M.,* ***Visioni, D.*** *et al.*, Earth's Future, 14, e2025EF007303.

102. 2026 **Projected future of African Marine ecosystems under climate change and stratospheric aerosol injection**, *Awo, F. M., Abiodun, B. J., Ansorge, I., Tomety, F. S., Dimoune, D. M., Samuelsen, A.,* ***Visioni, D.*** *et al.*, Journal of Geophysical Research: Oceans, 131, e2025JC022687.

101. 2026 **Stratospheric aerosol injection geoengineering has the potential to increase land carbon storage and to protect the Amazon rainforest**, *Parry, I. M., Ritchie, P. D. L., Boucher, O., Cox, P. M., Haywood, J. M., Niemeier, U., Seferian, R., Tilmes, S., and* ***Visioni, D.***, Earth Syst. Dynam., 17, 387–414, https://doi.org/10.5194/esd-17-387-2026.

100. 2026 **The global climate response to High-Latitude Low-Altitude Stratospheric Aerosol Injection (HiLLA-SAI)**, *A. Duffey, A., Lee, W., Wheeler, L., Irvine, P., Wagman, B., Henry, M.,* ***Visioni, D.***, *Tsamados, M., and MacMartin, D.*, Earth Syst. Dynam., 17, 353–385, https://doi.org/10.5194/esd-17-353-2026.

99. 2026 **Radiative forcing–based accounting (RFA): a dynamic framework to replace static GWPs in carbon markets**, *Agarwal, S., Srivastava, S., Kapoor, B., Osho, Z. and* ***Visioni, D.***, Environ. Res. Letters, 21 064024.

98. 2026 **A Climate Intervention Dynamical Emulator (CIDER) for scenario space exploration**, *Farley, J., MacMartin, D. G.,* ***Visioni, D.***, *Kravitz, B., Bednarz, E. M., Duffey, A., Henry, M., and Akherati, A.*, Geosci. Model Dev., 19, 1809–1831, https://doi.org/10.5194/gmd-19-1809-2026.

97. 2026 **South America Monsoon Lifecycle under SAI and no-SAI scenarios in a warming world**, *Ribeiro J. G. M., Reboita, M. S. and* ***Visioni, D.***, Environ. Res.: Climate 5 015023, DOI:10.1088/2752-5295/ae42df.

96. 2026 **Uncertainties of SAI efficiency and impacts depending on the complexity of the aerosol microphysical model**, *Tilmes, S.,* ***Visioni, D.***, *Quaglia, I., Zhu, Y., Bardeen, C. G., Vitt, F., and Yu, P.*, Atmos. Chem. Phys., 26, 2649–2666, https://doi.org/10.5194/acp-26-2649-2026, 2026.

95. 2026 **Sensitivity of Arctic sea ice recovery to stratospheric aerosol injection latitude**, *Kim, H., Kim, H.,* ***Visioni, D.***, *Bednarz, E. M.*, npj Clim Atmos Sci 9, 24 https://doi.org/10.1038/s41612-025-01298-0.

94. 2026 **Potential impacts of climate interventions on marine ecosystems**, *Roberts, K. E.,* ***Visioni, D.***, *Rohr, T., Raven, M. R., Diamond, M. S., Kravitz, B., et al.*, Reviews of Geophysics, 64, e2024RG000876.

93. 2026 **Air quality impacts of Stratospheric Aerosol Injections are small and mainly driven by changes in climate, not deposition**, *Wang, C.,* ***Visioni, D.***, *Chua, G., and Bednarz, E. M.*, Atmos. Chem. Phys., 26, 1339–1357, https://doi.org/10.5194/acp-26-1339-2026.

92. 2025 **Response of tipping elements to different strategies of stratospheric aerosol injection**, *Zhao, M., Cao, L.,* ***Visioni, D.***, *MacMartin, D. G.*, Earth's Future, 13, e2025EF006736. https://doi.org/10.1029/2025EF006736.

91. 2025 **Global and regional thermosteric and dynamic sea level change under stratospheric aerosol injection**, *Bonou, F., et al.*, Environ. Res.: Climate 4 045013, DOI 10.1088/2752-5295/ae15b6.

90. 2025 **Multi-model future world aridity and groundwater recharge changes with and without stratospheric aerosol intervention under high warming scenario**, *Rezaei, A., Moore, J., Tilmes, S., **Visioni, D.**, and Hussain, A.*, Geophysical Research Letters, 52, e2025GL117234.

89. 2025 **Hunga Tonga–Hunga Ha apai Volcano Impact Model Observation Comparison (HTHH-MOC) project: experiment protocol and model descriptions**, *Zhu, Y., et al.*, Geosci. Model Dev., 18, 5487–5512, https://doi.org/10.5194/gmd-18-5487-2025.

88. 2025 **Stratospheric Aerosol Injection Could Prevent Future Atlantic Meridional Overturning Circulation Decline, But Injection Location is Key**, *Bednarz, E. M., Goddard, P. B., MacMartin, D. G., **Visioni, D.**, Bailey, D., and Danabasoglu, G.*, Earth's Future, 13, e2025EF005919.

87. 2025 **Using optimization tools to explore stratospheric aerosol injection strategies**, *Brody, E., Zhang, Y., MacMartin, D. G., **Visioni, D.**, Kravitz, B., and Bednarz, E. M.*, Earth Syst. Dynam., 16, 1325–1341, https://doi.org/10.5194/esd-16-1325-2025.

86. 2025 **Assessing regional climate trends in West Africa under geoengineering: A multimodel comparison of UKESM1 and CESM2**, *Nkrumah, F., Quagraine, K. A., Quenum, G. M. L. D., **Visioni, D.**, Koffi, H. A., Klutse, N. A. B.*, Journal of Geophysical Research: Atmospheres, 130, e2024JD043117. https://doi.org/10.1029/2024JD043117.

85. 2025 **Key gaps in models' physical representation of climate intervention and its impacts**, *Eastham, S. D., Butler, A. H., Doherty, S. J., Gasparini, B., Tilmes, S., Bednarz, E. M., **Visioni, D.** et al.*, Journal of Advances in Modeling Earth Systems, 17, e2024MS004872. https://doi.org/10.1029/2024MS004872.

84. 2025 **Stratospheric Aerosol Intervention experiment for the Chemistry–Climate Model Initiative**, *Tilmes, S., Bednarz, E. M., Jörimann, A., **Visioni, D.**, Kinnison, D. E., Chiodo, G., and Plummer, D.*, Atmos. Chem. Phys., 25, 6001–6023, https://doi.org/10.5194/acp-25-6001-2025.

83. 2025 **Models and scenarios for solar radiation modification need to include human perceptions of risk**, *Beckage, B., Lacasse, K., Raimi, K. T., **Visioni, D.***, Environmental Research: Climate, 4, 023003, https://doi.org/10.1088/2752-5295/addd42.

82. 2025 **How marine cloud brightening could also affect stratospheric ozone**, *Bednarz, E., Haywood, J. M., **Visioni, D.**, Butler, A. H., Jones, A.*, Science Advances, 11, eadu4038(2025). DOI:10.1126/sciadv.adu4038.

81. 2025 **Baseline Climate Variables for Earth System Modelling**, *Juckes, M., Taylor, K. E., Antonio, F., Brayshaw, D., Buontempo, C., Cao, J., Durack, P. J., Kawamiya, M., Kim, H., Lovato, T., Mackallah, C., Mizielinski, M., Nuzzo, A., Stockhause, M., **Visioni, D.**, Walton, J., Turner, B., O'Rourke, E., and Dingley, B.*, Geosci. Model Dev., 18, 2639–2663, https://doi.org/10.5194/gmd-18-2639-2025, 2025.

80. 2025 **Volcanically forced Madden–Julian oscillation triggers the immediate onset of El Nino**, *Kim, H., Min, S., Kim, D., **Visioni, D.***, Nature Communications, 16, 1399. https://doi.org/10.1038/s41467-025-56692-2.

79. 2025 **World Climate Research Programme lighthouse activity: an assessment of major research gaps in solar radiation modification research**, *Haywood, J. M., Boucher O., Lennard C., Storelvmo T., Tilmes S., **Visioni, D.***, Frontiers in Climate, 7, doi:10.3389/fclim.2025.1507479.

78. 2025 **The Science of Solar Radiation Modification: Stratospheric Aerosol Injections and Marine Cloud Brightening**, *__Visioni, D.__, Narenpitak, P., Honegger, M.*, Encyclopedia of Atmospheric Sciences, https://doi.org/10.1016/B978-0-323-96026-7.00170-3.

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77. 2024 **Informative risk analyses of radiative forcing geoengineering require proper counterfactuals**, *Lee, W.R., Diamond, M.S., Irvine, P., Reynolds, J. and* ***Visioni, D.***, Commun Earth Environ 5, 748 (2024). https://doi.org/10.1038/s43247-024-01881-y.

76. 2024 **Modeling 2020 regulatory changes in international shipping emissions helps explain anomalous 2023 warming**, *Quaglia, I. and* ***Visioni, D.***, Earth Syst. Dynam., 15, 1527–1541, https://doi.org/10.5194/esd-15-1527-2024.

75. 2024 **Emulating inconsistencies in stratospheric aerosol injection**, *Farley, J., MacMartin, D.G.,* ***Visioni, D.***, *Kravitz, B.*, Environ. Res.: Climate 3, 035012. https://doi.org/10.1088/2752-5295/ad519c.

74. 2024 **Research criteria towards an interdisciplinary Stratospheric Aerosol Intervention assessment**, *Simone Tilmes, Karen H Rosenlof,* ***Daniele Visioni***, *Ewa M Bednarz, Tyler Felgenhauer, Wake Smith, Chris Lennard, Michael S Diamond, Matthew Henry, Cheryl S Harrison, Chelsea Thompson*, Oxford Open Climate Change, Volume 4, Issue 1, 2024, kgae010, https://doi.org/10.1093/oxfclm/kgae010.

73. 2024 **Kicking the can down the road: understanding the effects of delaying the deployment of stratospheric aerosol injection**, *Brody, E.,* ***Visioni, D.***, *Bednarz, E.M., Kravitz, B., MacMartin, D.G., Richter, J.H., Tye, M.R.*, Environ. Res.: Climate 3, 035011. https://doi.org/10.1088/2752-5295/ad53f3.

72. 2024 **Multi-Model Simulation of Solar Geoengineering Indicates Avoidable Destabilization of the West Antarctic Ice Sheet**, *Moore, J. C., Yue, C., Chen, Y., Jevrejeva, S.,* ***Visioni, D.***, *Uotila, P., and Zhao, L.*, Earth's Future, 12, e2024EF004424. https://doi.org/10.1029/2024EF004424.

71. 2024 **Carbon cycle response to stratospheric aerosol injection with multiple temperature stabilization targets and strategies**, *Zhao, M., Cao, L.,* ***Visioni, D.***, *and MacMartin, D. G.* , Earth's Future, 12, e2024EF004474. https://doi.org/10.1029/2024EF004474.

70. **2024** **A Living Assessment of Different Materials for Stratospheric Aerosol Injection—Building Bridges Between Model World and the Messiness of Reality**, ***Visioni, D.***, *Quaglia, I., and Steinke, I.*, Geophysical Research Letters, 51, e2024GL108314. https://doi.org/10.1029/2024GL108314.

69. 2024 **Analysis of the global atmospheric background sulfur budget in a multi-model framework**, *Brodowsky, C. V., Sukhodolov, T., Chiodo, G., Aquila, V., Bekki, S., Dhomse, S. S., Höpfner, M., Laakso, A., Mann, G. W., Niemeier, U., Pitari, G., Quaglia, I., Rozanov, E., Schmidt, A., Sekiya, T., Tilmes, S., Timmreck, C., Vattioni, S.,* ***Visioni, D.***, *Yu, P., Zhu, Y., and Peter, T.*, Atmos. Chem. Phys., 24, 5513–5548, https://doi.org/10.5194/acp-24-5513-2024.

68. 2024 **Dependency of the impacts of geoengineering on the stratospheric sulfur injection strategy – Part 2: How changes in the hydrological cycle depend on the injection rate and model used**, *Laakso, A.,* ***Visioni, D.***, *Niemeier, U., Tilmes, S., and Kokkola, H.*, Earth Syst. Dynam., 15, 405–427, https://doi.org/10.5194/esd-15-405-2024.

67. 2024 **The potential of Stratospheric Aerosol Injection to reduce the climatic risks of explosive volcanic eruptions**, *Quaglia, I.,* ***Visioni, D.***, *Bednarz, E. M., MacMartin, D. G., and Kravitz, B.*, Geophysical Research Letters, 51, e2023GL107702. https://doi.org/10.1029/2023GL107702.

66. **2024** **G6-1.5K-SAI: a new Geoengineering Model Intercomparison Project (GeoMIP) experiment integrating recent advances in solar radiation modification studies**, ***Visioni, D.***, *Robock, A., Haywood, J., Henry, M., Tilmes, S., MacMartin, D. G., Kravitz, B., Doherty, S. J., Moore, J., Lennard, C., Watanabe, S., Muri, H., Niemeier, U., Boucher, O., Syed, A., Egbebiyi, T. S., Séférian, R., and Quaglia, I.*, Geosci. Model Dev., 17, 2583–2596, https://doi.org/10.5194/gmd-17-2583-2024.

65. 2024 **Hemispherically symmetric strategies for stratospheric aerosol injection**, *Zhang, Y., MacMartin, D. G., **Visioni, D.**, Bednarz, E. M., and Kravitz, B.*, Earth Syst. Dynam., 15, 191–213, https://doi.org/10.5194/esd-15-191-2024.

64. 2024 **Side Effects of Sulfur-Based Geoengineering Due To Absorptivity of Sulfate Aerosols**, *Wunderlin, E., Chiodo, G., Sukhodolov, T., Vattioni, S., **Visioni, D.**, and Tilmes, S.*, Geophysical Research Letters, 51, e2023GL107285.

63. 2023 **Potential Non-Linearities in the High Latitude Circulation and Ozone Response to Stratospheric Aerosol Injection**, *Bednarz, E. M.,**Visioni, D.**, Butler, A. H., Kravitz, B., MacMartin, D. G., and Tilmes, S.*, Geophysical Research Letters, 50, e2023GL104726.

62. 2023 **Stratospheric aerosol injection can reduce risks to Antarctic ice loss depending on injection location and amount**, *Goddard, P. B., Kravitz, B., MacMartin, D. G., **Visioni, D.**, Bednarz, E. M., and Lee, W. R.*, Journal of Geophysical Research: Atmospheres, 128, e2023JD039434.

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4. 2022 **Process-Level Experiments and Policy-Relevant Scenarios in Future GeoMIP Iterations**, ***Visioni, D.****, Robock, A., Duffey, A. and Quaglia, I.*, Bulletin of the American Meteorological Society, 104, E501-E503.

3. 2022 **Future geoengineering scenarios: balancing policy relevance and scientific significance**, ***Visioni, D.****, and Robock, A.*, Bulletin of the American Meteorological Society, 103(3), E817-E820.

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Teaching and mentorship activities

2025 **Instructor**, *Theory and Practice of Earth System Modeling*, EAS, Cornell University.
Developed new course, mainly at the graduate level, including practical examples for running CESM2

2025 **Instructor**, *Aerosols and Climate*, EAS, Cornell University.
Developed new course, mainly at the graduate level, using up-to-date research and discussion sections

2023-2025 **Instructor**, *Climate Dynamics*, Department of Earth and Atmospheric Sciences, Cornell University.
Capstone class for our department, with over 50 students per year (including incoming graduate students)

2021-Current **External examiner for PhD and Master thesis**.
Cambridge University (PhD); ETH Zurich (Master)

Sept 2020-2023 **LeadTheFuture STEM Mentorship Program**, *LeadTheFuture*.
Mentoring Italian Bachelor and Master students in STEM programs

Aug 2019-2022 **GSMU Mentorship Program**, Cornell University.
Mentoring first generation college students with an interest in pursuing a PhD

2018 **Lecturer**, *Atmospheric radiative transfer*, Department of Physical and Chemical Sciences, University of L'Aquila.

2017,2018 **Lecturer**, *Magnetism and Electricity Lab*, Department of Physical and Chemical Sciences, University of L'Aquila.

2017,2018 **Lecturer**, *General physics*, Department of Biology and Life Sciences, University of L'Aquila.

International conferences, talks and workshops

Attended as invited speaker

March 25, 2026 **2026 SCI-EX event**, *"Steps towards a more robust assessment of Climate Intervention Strategies"*, University of Texas at Austin, TX.

March 16, 2026 **Symposium on Temperature Overshoot**, *"Solar Radiation Modification impacts on the climate system in the context of overshoot"*, University of Tokyo, Tokyo.

April 25, 2025 **Eye Towards the Sky Annual Speaker Series: Human Ingenuity vs Climate Change**, `https://eyetowardsthesky.com/`, Boston, MA.

March 14, 2025 **2025 Isaac Asimov Memorial Debate: the Promises and Pitfalls of Geoengineering**, Youtube recording, American Museum of Natural History, NY.

November 4, 2024 **Department of Environmental Sciences Seminar Series**, *"Climate Intervention with Stratospheric Aerosols: steps towards a robust assessment"*, University of Virginia.

October 17, 2024 **Solar Geoengineering Research Program Lunch Talk**, *"Climate Intervention with Stratospheric Aerosols: steps towards a robust assessment"*, YouTube recording, Harvard, MA.

February 22, 2024 **Gordon Research Conference on Climate Engineering**, *"Keynote address: Climate Modeling for Sunlight Reflection Methods: Progress So Far, Why We Do It, Who Is it For, and What Is Next "*, Florence, Italy.

October 20, 2023 **College of the Coast and Environment (CCE) Seminar at LSU**, *"Climate Intervention with Stratospheric Aerosols: steps towards a more robust assessment"*, Baton Rouge, LA, USA.

July 12, 2023 **28th IUGG General Assembly**, *"The Geoengineering Model Intercomparison Project (GeoMIP): Past, present and future"*, Berlin, Germany.

November 15, 2022 **Climate and Global Dynamics seminar series, NCAR**, *"Stratospheric aerosol geoengineering: How do we move towards a more robust assessment"*, Boulder, CO, USA, YouTube recording.

September 23, 2022 **Lamont-Doherty Earth Observatory Earth Science Colloquium**, *"Understanding the potential impacts of sulfate aerosol injections on the climate system"*, Columbia University, New York, NY, USA, event link.

June 28-July 3, 2022 **Gordon Research Conference on Climate Engineering**, *"The Role of Different Temperature Targets for Determining Climate Engineering Outcomes and Trade-Offs"*, Sunday River-Newry, ME, USA.

April 11th 2022 **Atmospheric Chemistry Observation and Modeling Virtual Seminar**, *"Stratospheric aerosol geoengineering: understanding and reducing modeling uncertainties"*, YouTube recording.

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March 22nd 2022 **Labont seminars: Climate Crisis and Future Generations, Sant'Anna School of Advanced Studies, Italy**, *"Climate engineering: what do we know, what do we still need to know, and should we know it?"*, YouTube recording.

January 13th 2022 **University of Perugia, Italy, Department of Physics and Geology**, *"The Sun-Earth relationship across different timescales and its climatic influences"*.

January 10th 2022 **University of Washington, Department of Atmospheric Sciences**, *"Understanding the potential impacts of sulfate aerosol injections on the climate system"*.

October 13th 2021 **Yale College**, *"Earth System Modeling applied to Geoengineering"*.
As part of the course Geoengineering: Climate Change held by Prof. W. Smith

August 12, 2021 **Center for Climate Repair at Cambridge Summer Workshop Series**, *"Refreezing the Arctic: Stratospheric Aerosol Injection and other techniques"*, Center for Climate Repair at Cambridge, Cambridge, UK, YouTube recording.

August 1-7, 2021 **Ecological Society of America Annual Meeting 2021**, *"What goes up must come down: surface impacts of deposition in a sulfate geoengineering scenario"*, Long Beach, California.

Jan 10-14, 2021 **American Meteorological Society Annual Meeting 2021**, *"Geoengineering with stratospheric aerosols - physical mechanisms and sources of uncertainty"*, American Meteorological Society, New Orleans, USA.

30 Sep 2019 **Geoengineering Modeling Research Consortium, 2nd meeting**, *"Comparison of SO2 and H2SO4 injection strategies using a model aerosol microphysics representation"*, Harvard University, Cambridge, MA, USA.

20-21 May 2019 **Geoengineering Modeling Research Consortium, 1st meeting**, *"Changes in sulfate geoengineering efficacy due to uncertainties in model representations of high clouds"*, NCAR, Boulder, CO, USA.

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